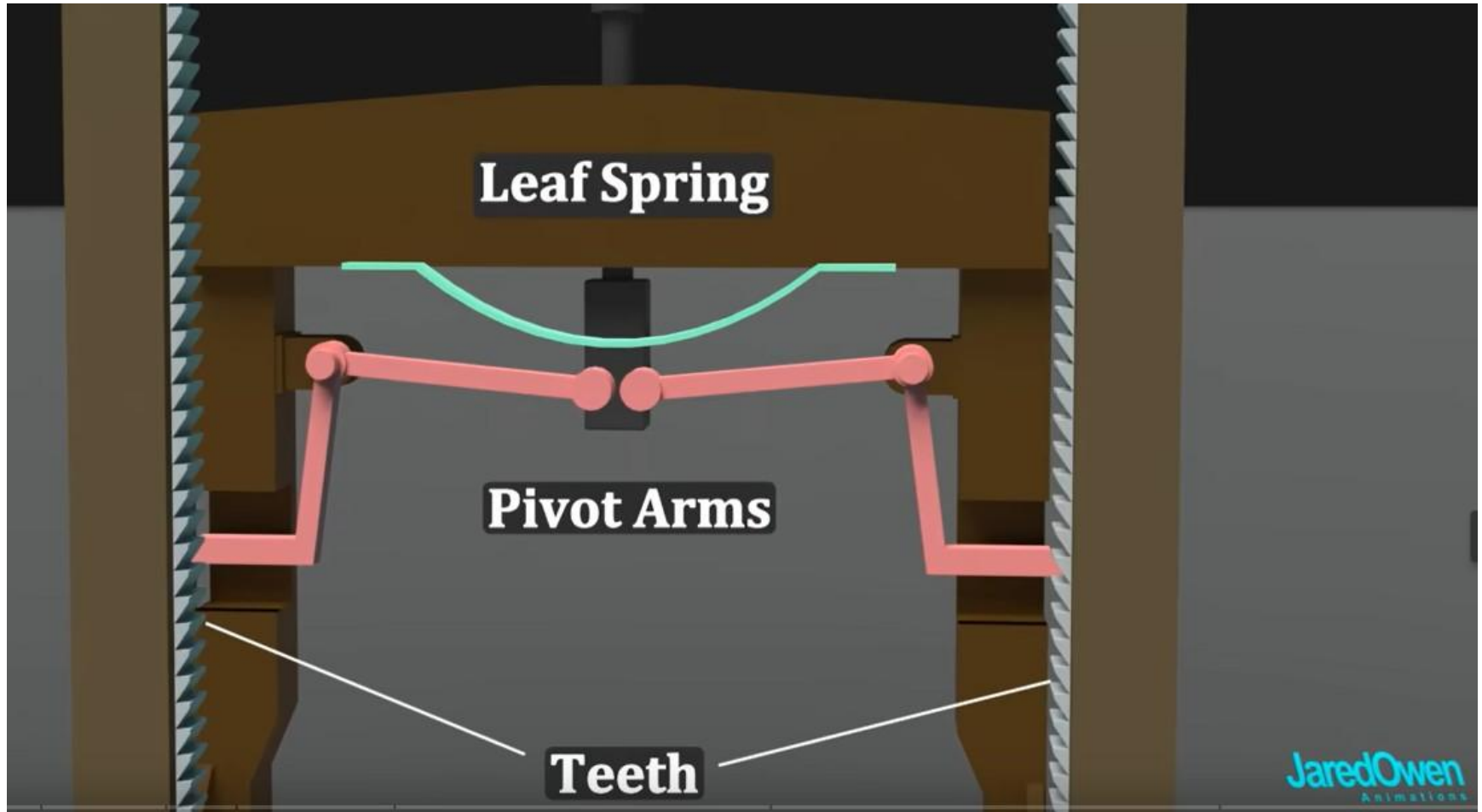


Elevator Safety Mechanism

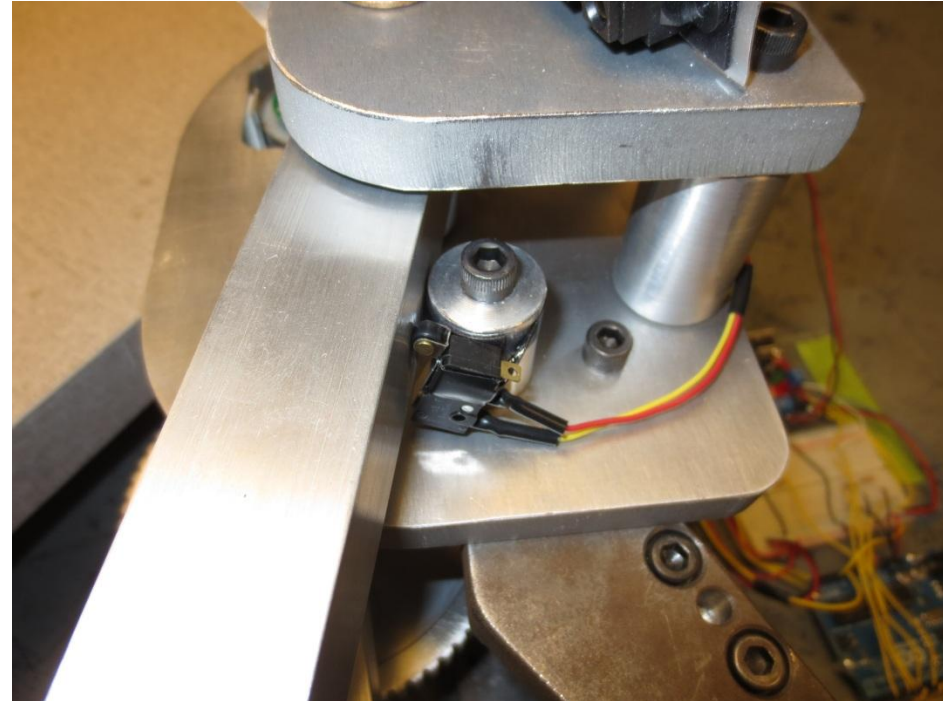


<https://www.youtube.com/watch?v=rKp4pe92ljg>

Sample parts to Pass Around

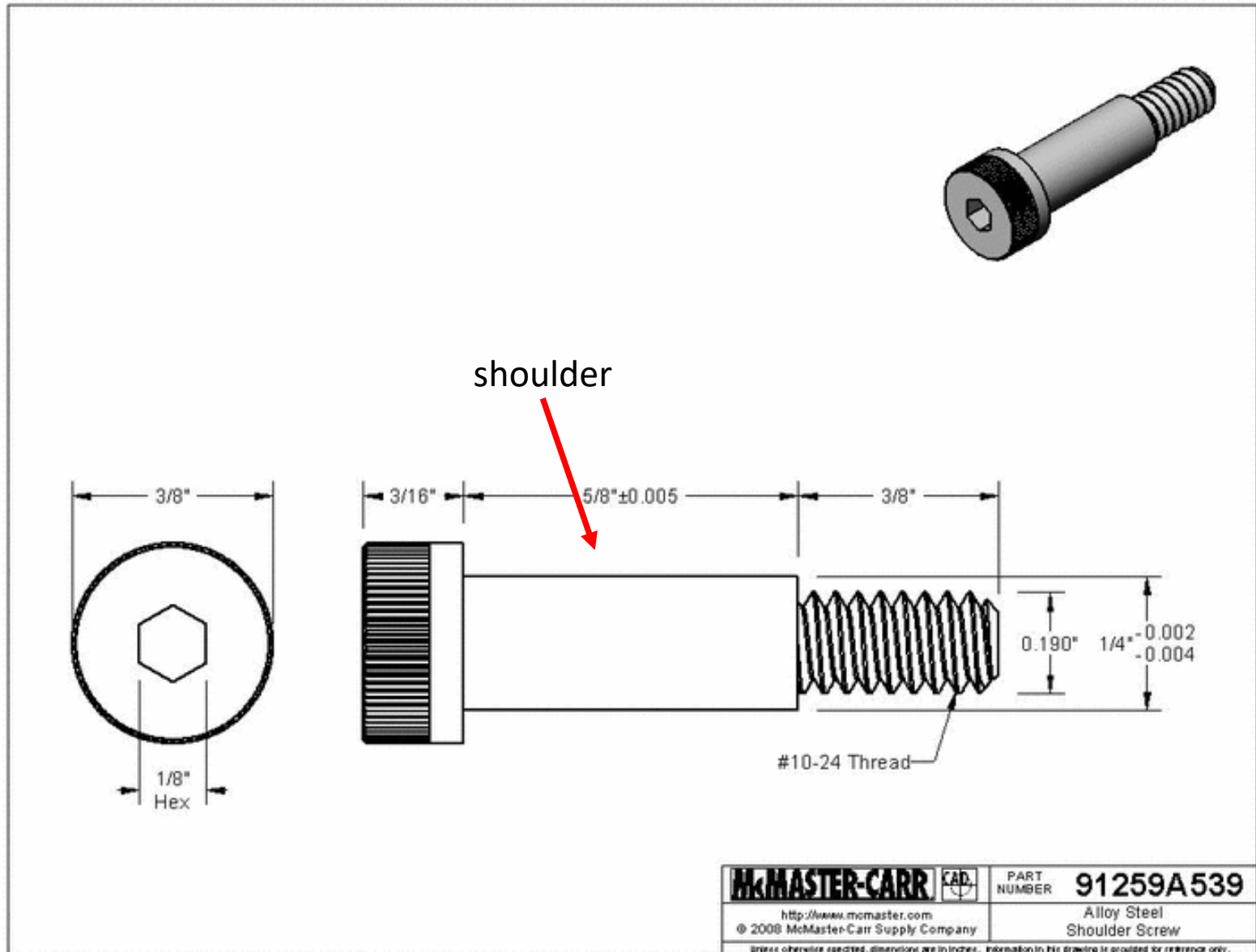
- Shoulder screw and bushing
- Example of transferring torque from a gear to a link
- Bent link

Hard Stop (Positive Stop)



A hard stop ensures that the linkage will come to a stop in a repeatable position each time. These pictures show one type of a hard stop, also with a limit switch to signal the linkage's position to the Arduino.

Shoulder screws



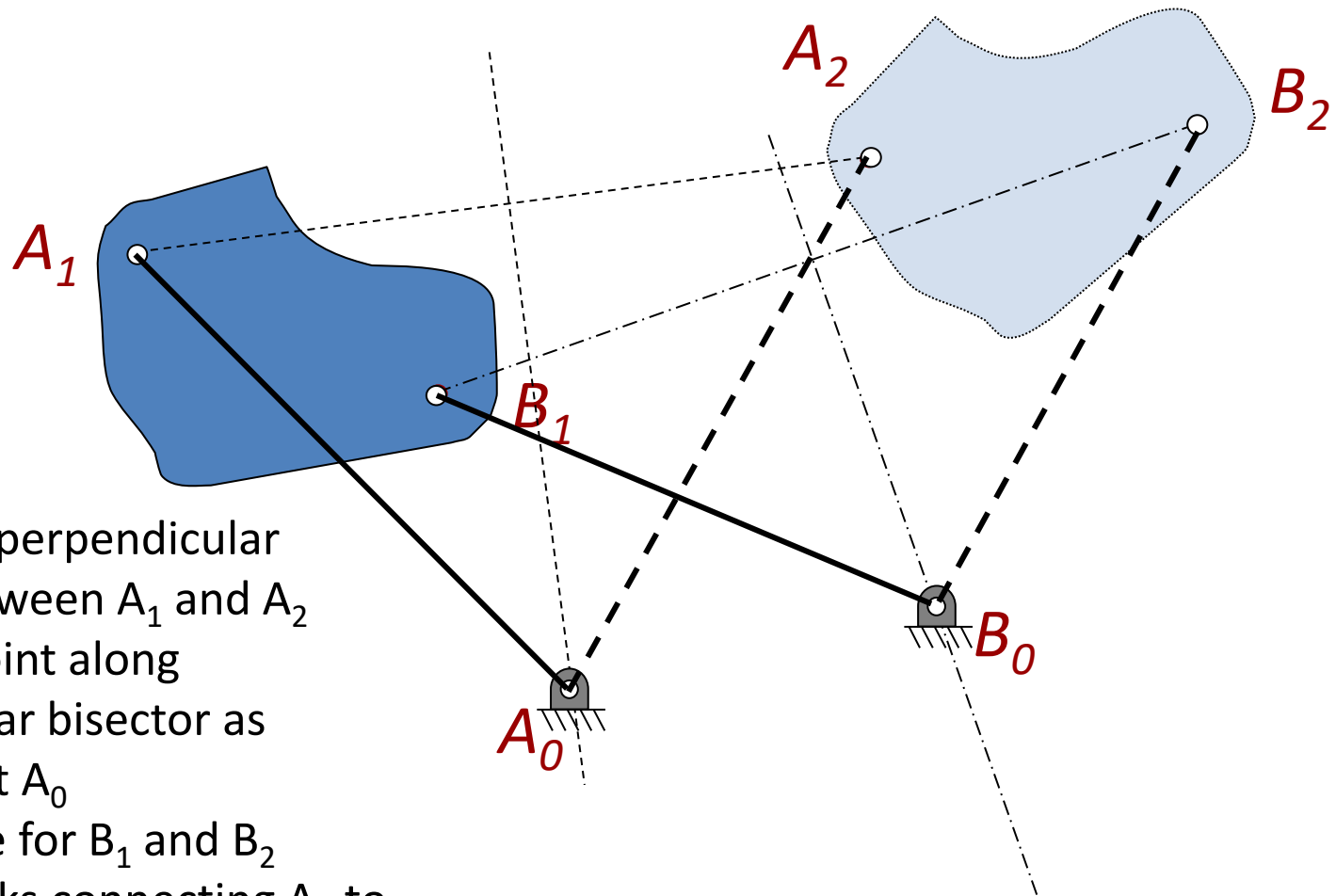
ME350 Software Subject Matter Experts (Gurus)

If you have a question about one of the applications that we use in ME350, you should first ask your lab section instructor.

For very in-depth questions, you can consult one of our gurus for ME350 software:

- SolidWorks: Haoran (haorshi) & James (jplorezn)
- ADAMS: Lawrence (chenlawr) & Moeen (moeenrad)
- Arduino: Nana (narmph) & Chloe (caffaki)

Graphical Method: Motion Generation with 2 Prescribed Positions

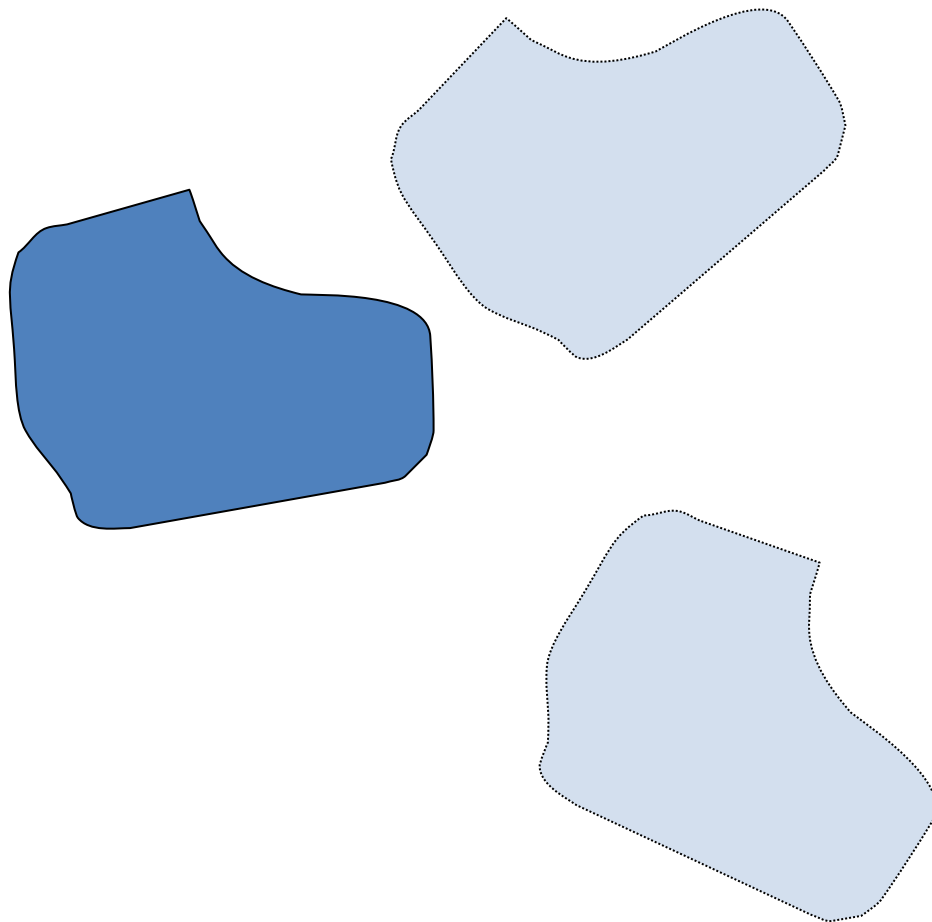


- Construct a perpendicular bisector between A_1 and A_2
- **Choose** a point along perpendicular bisector as ground pivot A_0
- Do the same for B_1 and B_2
- Draw the links connecting A_1 to A_0 and B_1 to B_0

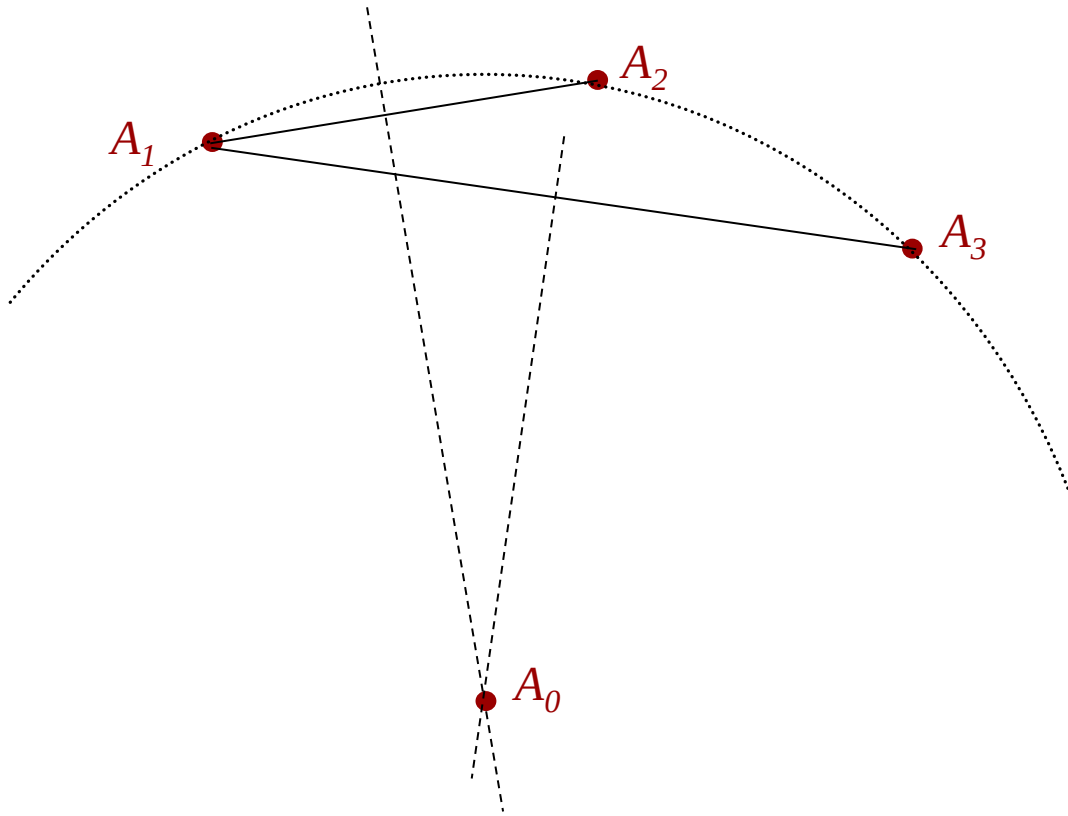
3-Pt Motion Generation (Graphical)

What if we have 3 specified positions?

- 3 non-collinear points in 2-D lie along an arc **with a particular radius**
- Find the ground pivots by constructing PBs

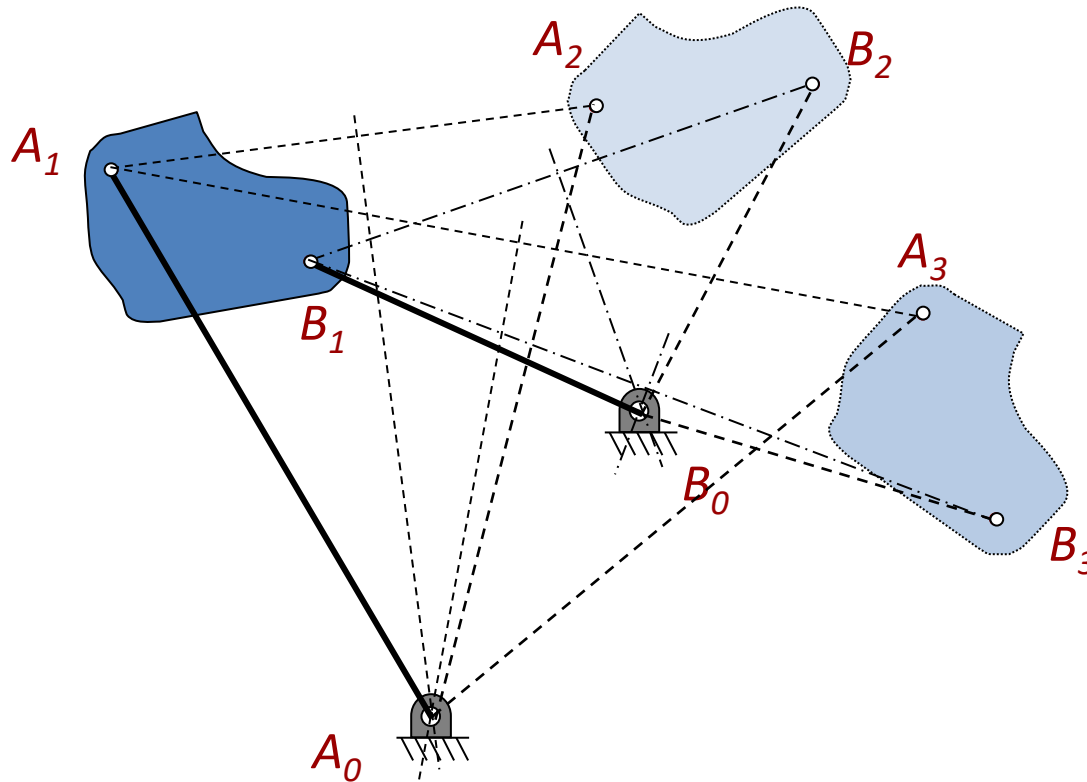


3-Pt Motion Generation (Graphical)



- Construct PBs of A_1A_2 and A_1A_3
- The center point (ground pivot) A_0 must lie along both lines.
 - A_0 is located at intersection of PBs.

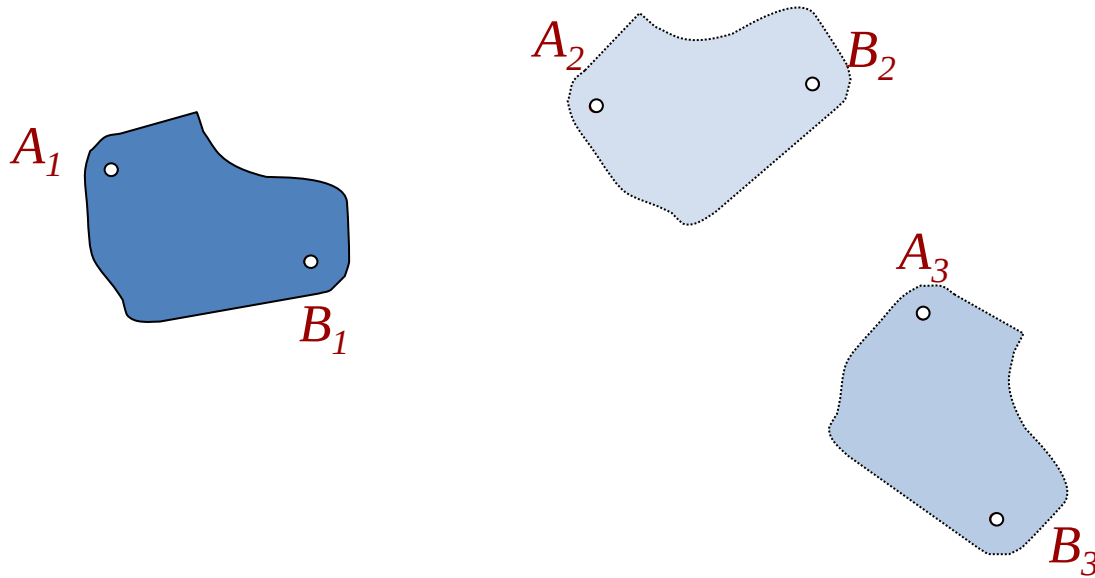
Graphical Method: Motion Generation with 3 Prescribed Positions

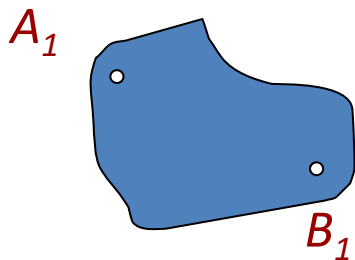


- Construct perpendicular bisectors. There are 2 for the A points and 2 for the B points.
- The intersection of the corresponding perpendicular bisectors locates the ground pivots.
- Draw in links connecting A_1 to A_0 and B_1 to B_0 .

When you use this graphical method, show your work. Do not erase your construction lines afterwards.

3-Pt Motion Generation (Graphical)



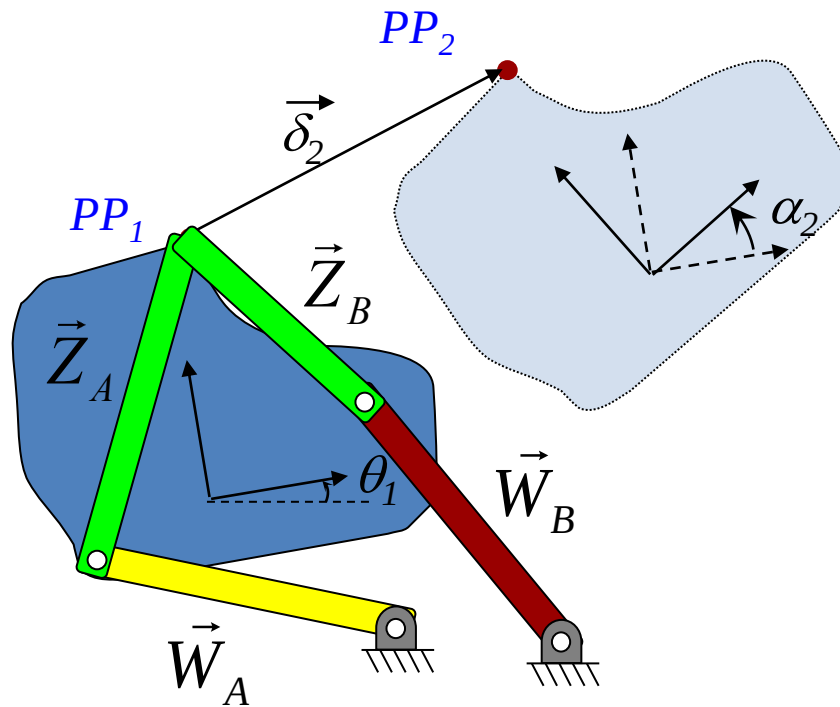


Graphical Synthesis: a few notes

- How many free choices do we have in finding solutions?
 - If 2 prescribed positions: 3 free choices for each side (A side, B side). 6 total
 - 2 free choices come from selecting the location (x and y coordinates) of the moving pivots on the rigid body that needs to be moved.
 - 1 free choice comes from selecting the ground pivots along the perpendicular bisector.
 - If 3 prescribed positions: 2 free choices for each side. 4 total
 - 2 free choices come from location (X and Y coordinates) of moving pivots.
- What are some challenges with graphical synthesis?
 - Everything depends on how straight and perpendicular the lines are.
 - Each iteration takes a relatively long time to do.
- How may we address these challenges?
 - Analytical (formulas) synthesis

What if there are only two prescribed positions instead of three?

There is only one equation for each dyad, instead of two:



$$\vec{W}(e^{i\beta_2} - 1) + \vec{Z}(e^{i\alpha_2} - 1) = \vec{\delta}_2$$

Knowns: $\alpha_2, \vec{\delta}_2$

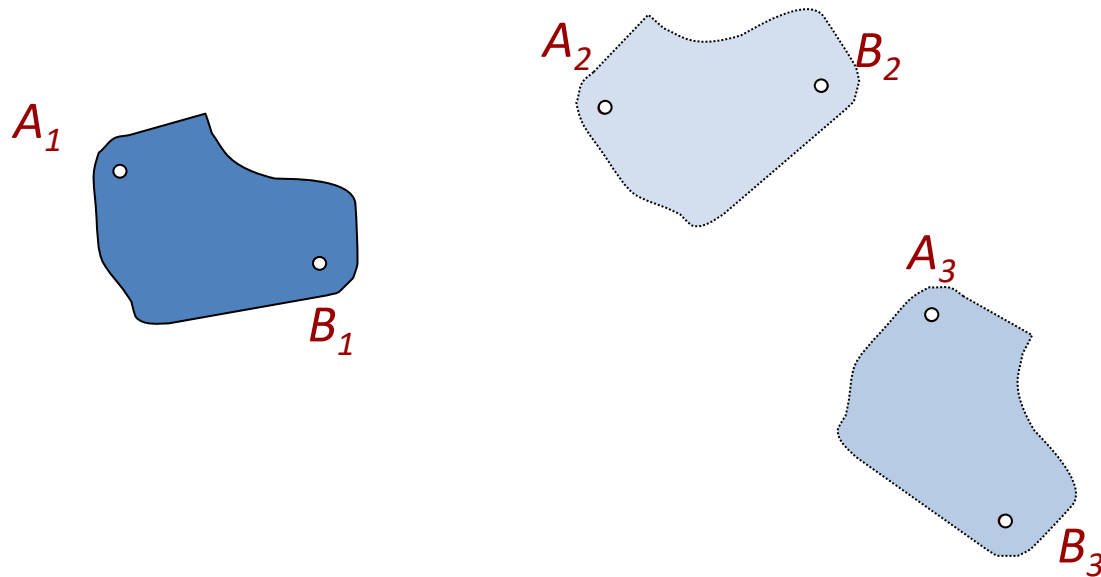
Free Choices: $\beta_2, \vec{Z}$

(“Moving Pivots Specification”)

Solve this equation for the $\vec{W}$ vector.

ME350 PUZZLE

How could you write the equations of motion of this part, in such a way that you could use a computer to help design a linkage to guide it through those positions?



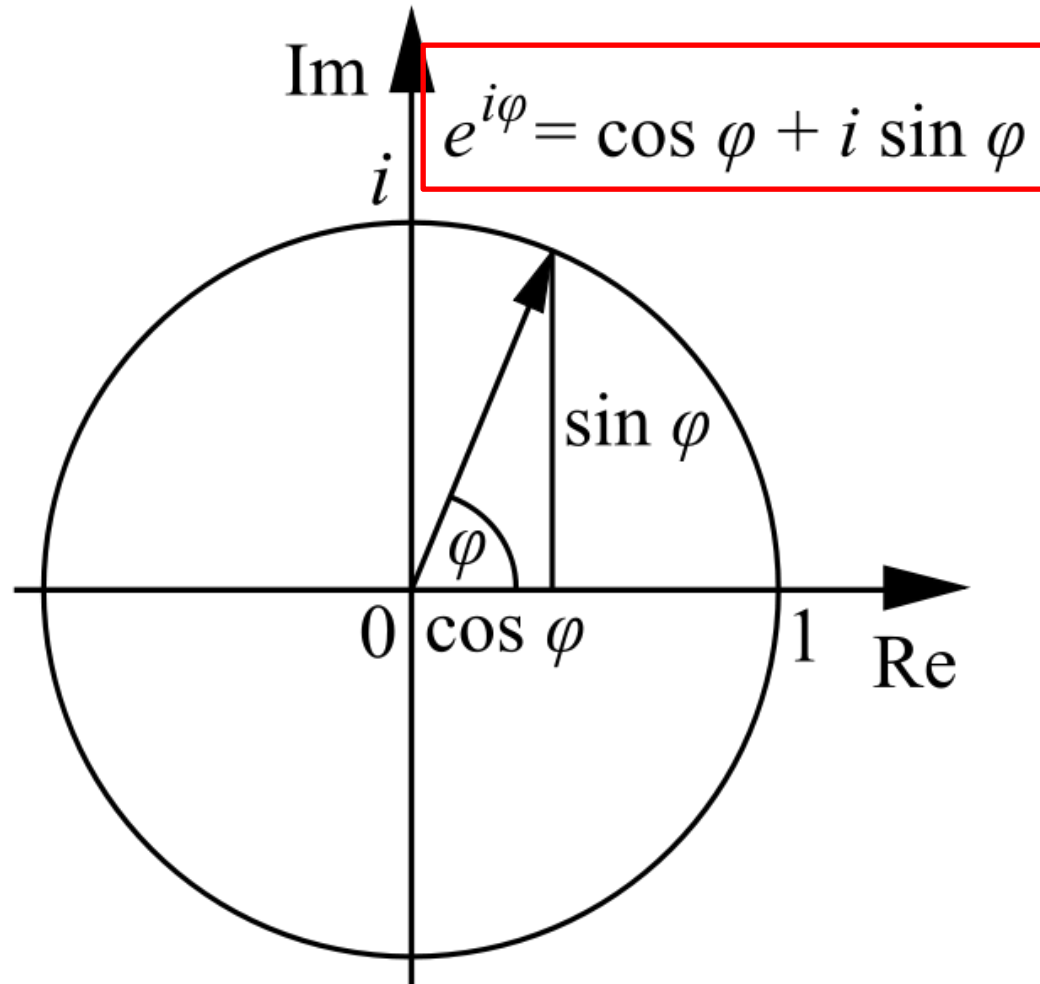
Complex Number Review

These concepts are used in analytical synthesis:

- Euler's formula and $e^{i\varphi}$
- Vectors: Polar Notation and Cartesian Notation
- Cramer's rule

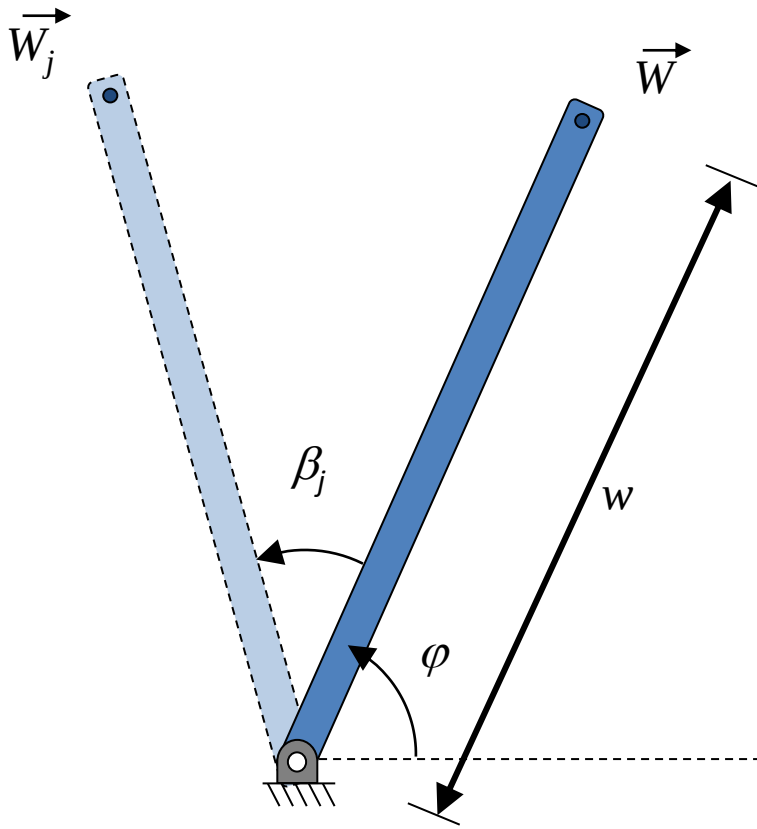
Euler's Formula

- Valid for any real number φ
- e : base of natural logarithm
- i : imaginary unit
 $i = \sqrt{-1}$



Vectors and Complex Numbers

Multiplying a vector by $e^{i\varphi}$ means rotate that vector by φ radians counterclockwise.



$$\vec{W} = we^{i\varphi}$$

$$\vec{W}_j = we^{i(\varphi + \beta_j)}$$

$$= we^{i\varphi} e^{i\beta_j}$$

$$= \vec{W} e^{i\beta_j}$$

Cramer's Rule

Cramer's Rule uses determinants

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

to solve a system of equations
where there are as many
unknowns as the number of
equations.



■ Example:

$$ax + by = e$$

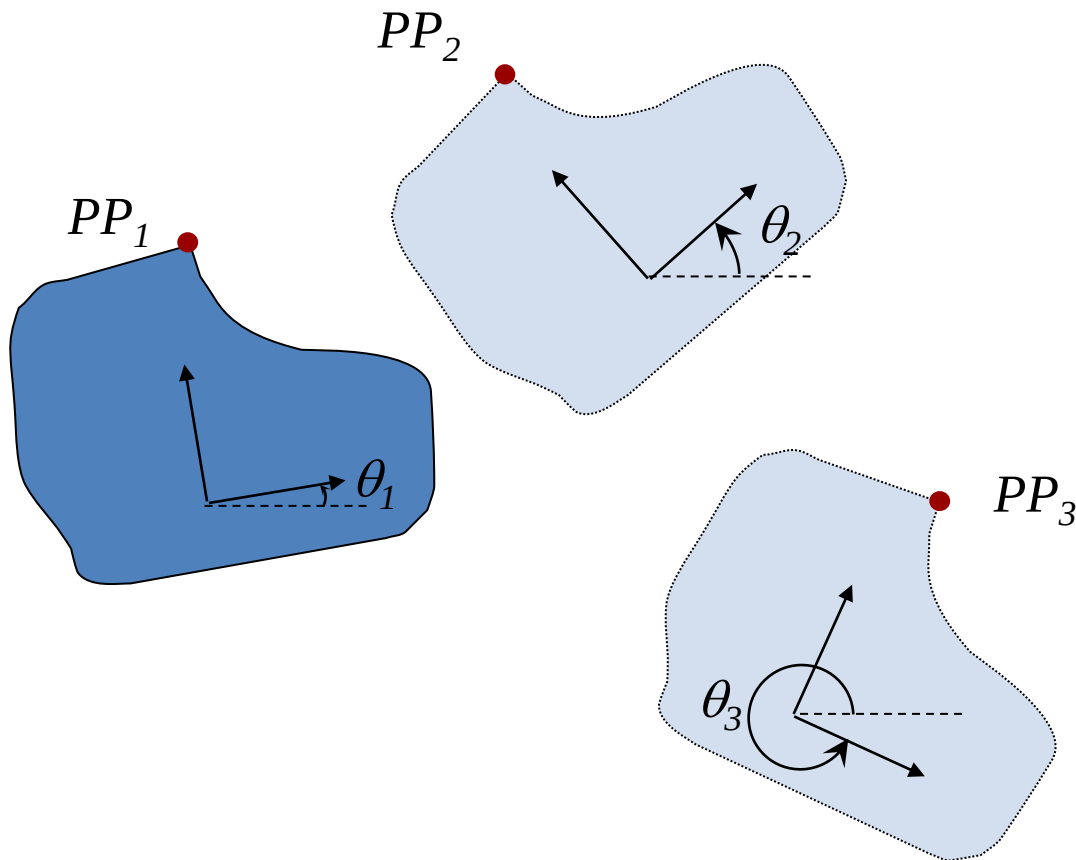
$$cx + dy = f$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} e \\ f \end{bmatrix}$$

$$x = \frac{\begin{vmatrix} e & b \\ f & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}} = \frac{ed - bf}{ad - bc}$$

$$y = \frac{\begin{vmatrix} a & e \\ c & f \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}} = \frac{af - ec}{ad - bc}$$

Problem Specification

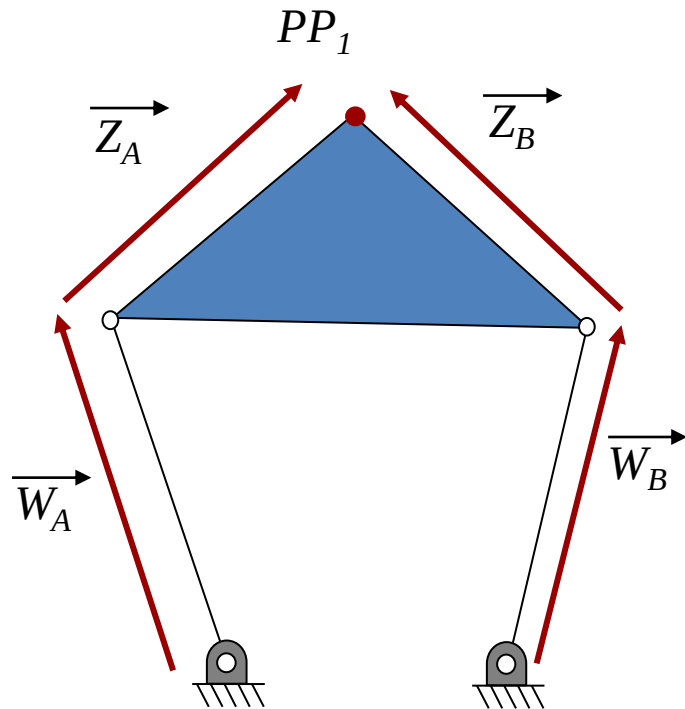


PP_j - location of a single point on coupler link in j^{th} position

Called a “precision point”

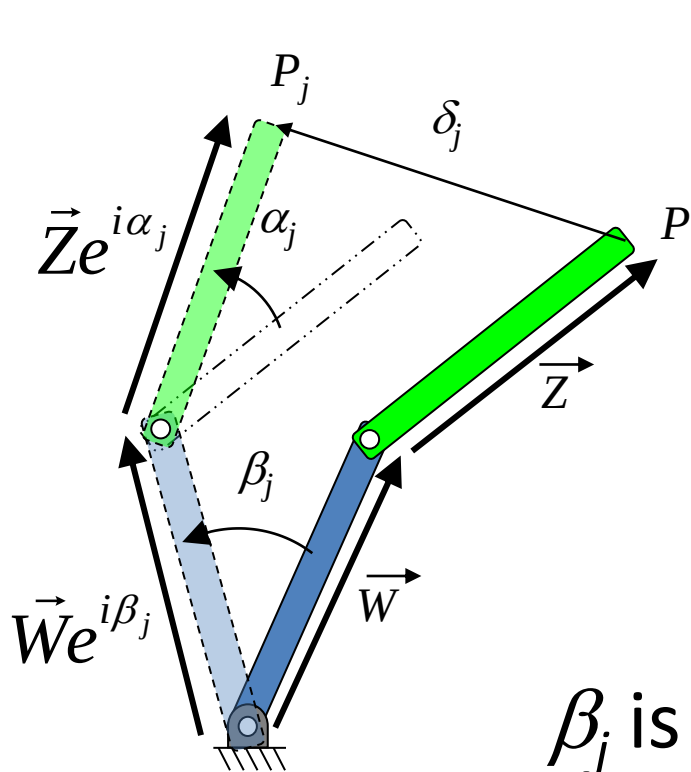
θ_j - orientation of coupler with respect to some fixed coordinate frame

Vector Representation for Standard Dyad Form



- A four bar linkage is modeled using two dyads.
 - A dyad is a combination of two vectors: a W vector and a Z vector together.
- The Z vectors are rigidly connected to each other.
 - They form the coupler.

Standard Dyad Form, and definition of β_j



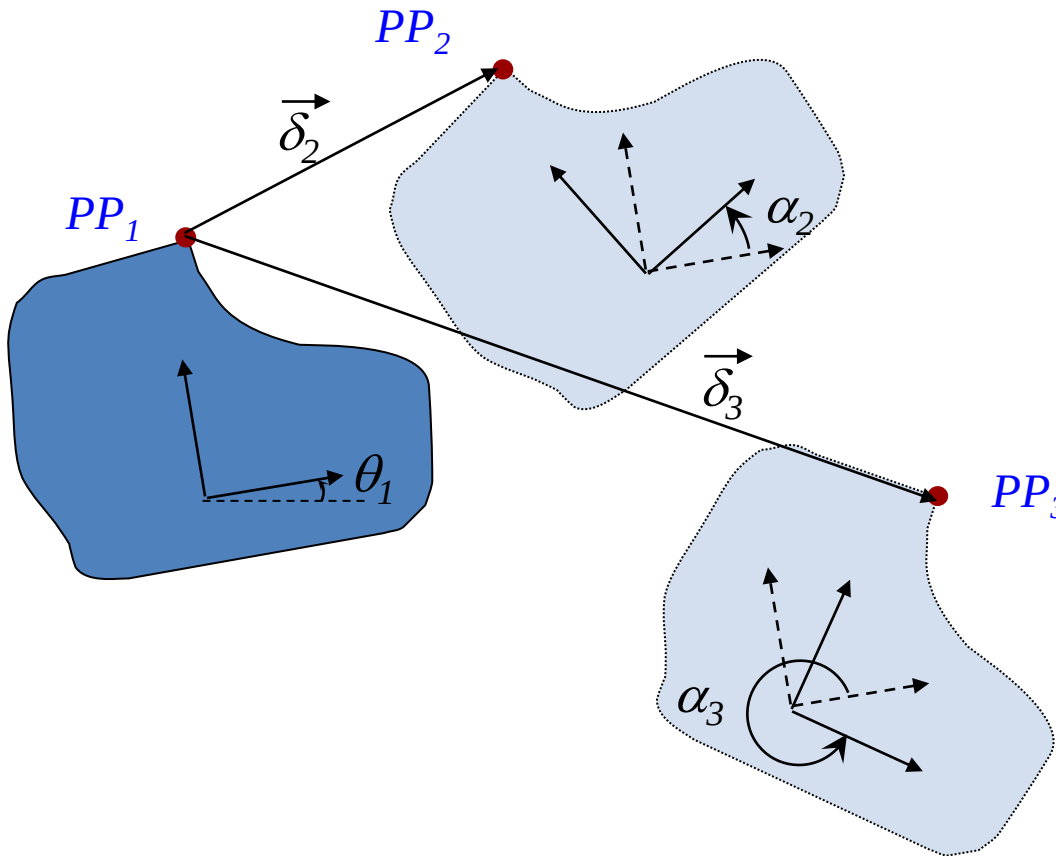
$$\vec{\delta}_j = P_j - P$$

$$= \vec{W}e^{i\beta_j} + \vec{Z}e^{i\alpha_j} - (\vec{W} + \vec{Z})$$

$$= \vec{W}(e^{i\beta_j} - 1) + \vec{Z}(e^{i\alpha_j} - 1)$$

β_j is the change in angle of the link that is connected to the ground.

Definitions: Delta (δ) and Alpha (α)



- δ_j is the vector from PP_1 to PP_j
 $\vec{\delta}_2 = \overrightarrow{PP_2 - PP_1}$
 $\vec{\delta}_3 = \overrightarrow{PP_3 - PP_1}$
- α_j is the change in orientation of coupler with respect to orientation in first position

$$\alpha_2 = \theta_2 - \theta_1$$

$$\alpha_3 = \theta_3 - \theta_1$$

Dyad Synthesis

How do we use the standard dyad form?

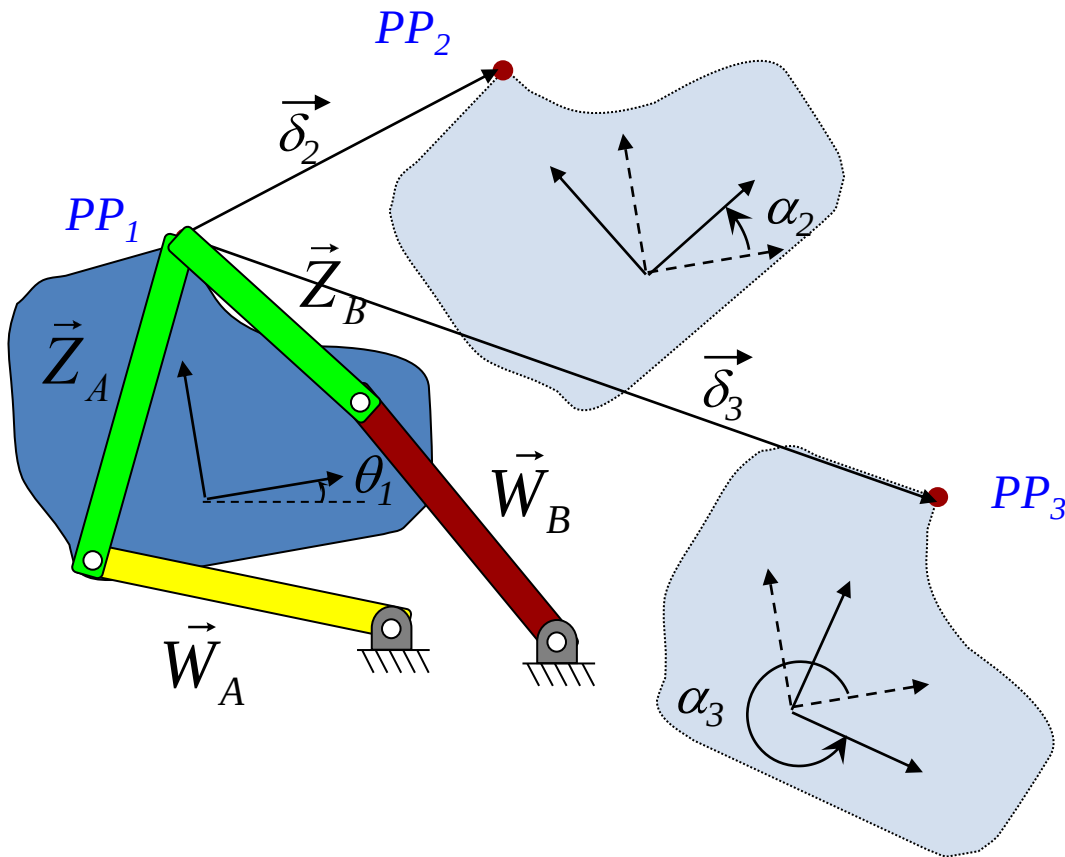
- Consider 3 Point Motion Generation

$$\vec{W} \left(e^{i\beta_2} - 1 \right) + \vec{Z} \left(e^{i\alpha_2} - 1 \right) = \vec{\delta}_2$$

$$\vec{W} \left(e^{i\beta_3} - 1 \right) + \vec{Z} \left(e^{i\alpha_3} - 1 \right) = \vec{\delta}_3$$

- We know $\alpha_2, \alpha_3, \delta_2$, and δ_3
- 6 unknowns
 - 2 vector: W and Z
 - 2 scalar: β_j
- 4 equations (2 vector)
 - 2 free choices
 - Typically, we choose β_j 's $\rightarrow$ Linear equations

Motion Generation for Three Precision Points



Knowns:

$$\vec{\delta}_j, \alpha_j$$

Unknowns:

$$\vec{W}, \vec{Z}, \beta_j$$

Analytical Method:

Motion Generation with 3 Prescribed Positions

Problem Formulation:

$$\mathbf{W}(e^{i\beta_2} - 1) + \mathbf{Z}(e^{i\alpha_2} - 1) = \delta_2$$

$$\mathbf{W}(e^{i\beta_3} - 1) + \mathbf{Z}(e^{i\alpha_3} - 1) = \delta_3$$

Solution:
(using
Cramer's Rule)

$$\mathbf{W} = \frac{\begin{vmatrix} \delta_2 & e^{i\alpha_2} - 1 \\ \delta_3 & e^{i\alpha_3} - 1 \end{vmatrix}}{\begin{vmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \end{vmatrix}}$$

$$\mathbf{Z} = \frac{\begin{vmatrix} e^{i\beta_2} - 1 & \delta_2 \\ e^{i\beta_3} - 1 & \delta_3 \end{vmatrix}}{\begin{vmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \end{vmatrix}}$$

Expressing the delta vectors in MATLAB

(remember this for Homework!)

Instead of using polar coordinates for the delta vectors, you can use Cartesian coordinates if they are given (or easy to measure.)

The output from MATLAB will always be in Cartesian coordinates. Thus you can plot the results in the same way.

Complex numbers

You can represent complex numbers in a very straightforward manner using built-in functions in Matlab. For example, you can express the complex number $0.2 + 3i$ in Matlab as:

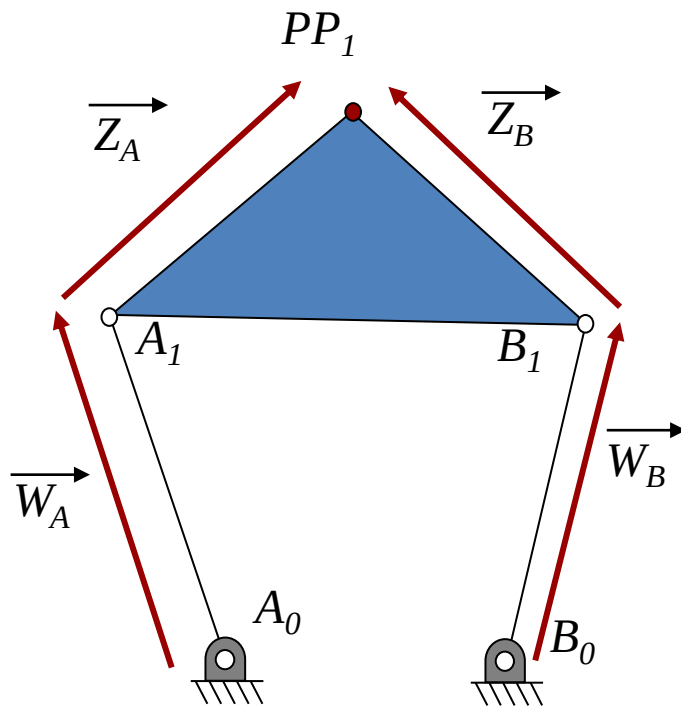
`0.2 + 3*i`

You may also want to express complex numbers using polar coordinates. For example, you can express the number $2e^{i56^\circ}$ in Matlab as:

`2*exp(i*56/180*pi)`

Determining the locations of the joints

Once we find $\vec{W}$ and $\vec{Z}$, how do we find the moving pivot and ground pivot, A_1 and A_0 ?



$$\vec{A_1} = \vec{PP_1} - \vec{Z_A}$$

$$\& \vec{A_0} = \vec{A_1} - \vec{W_A}$$

$$\vec{B_1} = \vec{PP_1} - \vec{Z_B}$$

$$\& \vec{B_0} = \vec{B_1} - \vec{W_B}$$

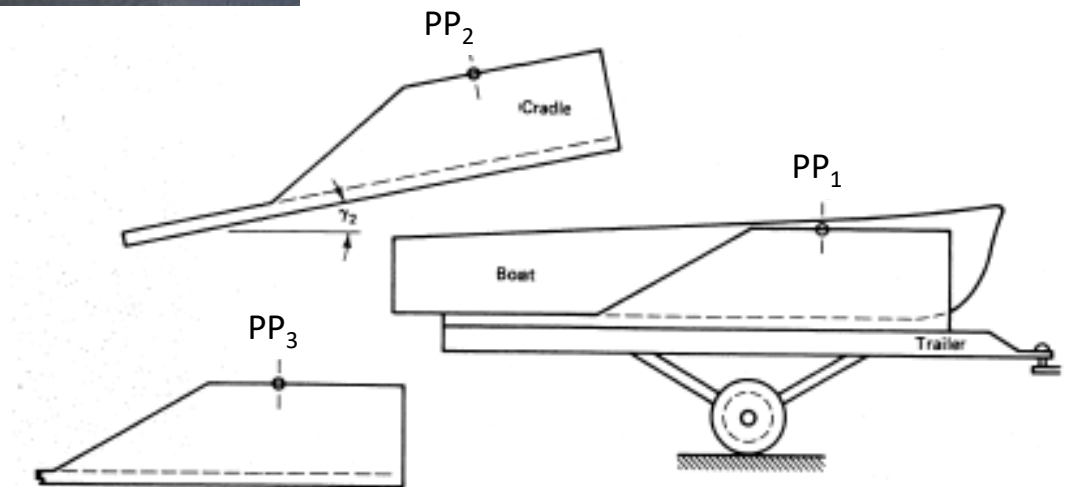
Then we do the same for B_1 and B_0 .

3-Pt Motion Generation Example

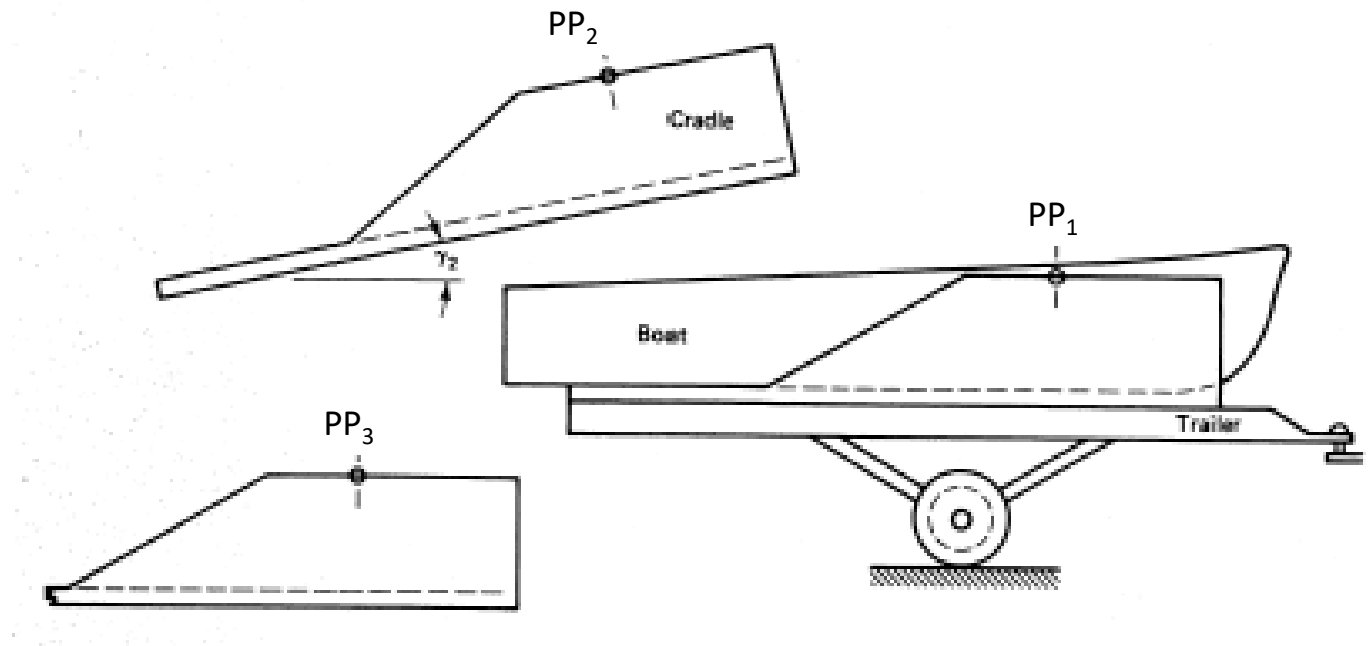


Start by measuring the delta and alpha values.

NOTE: Must convert from degrees to radians!



3-Pt Motion Generation Example



3-Pt Motion Generation Example

$$\alpha_2 = \theta_2 - \theta_1$$

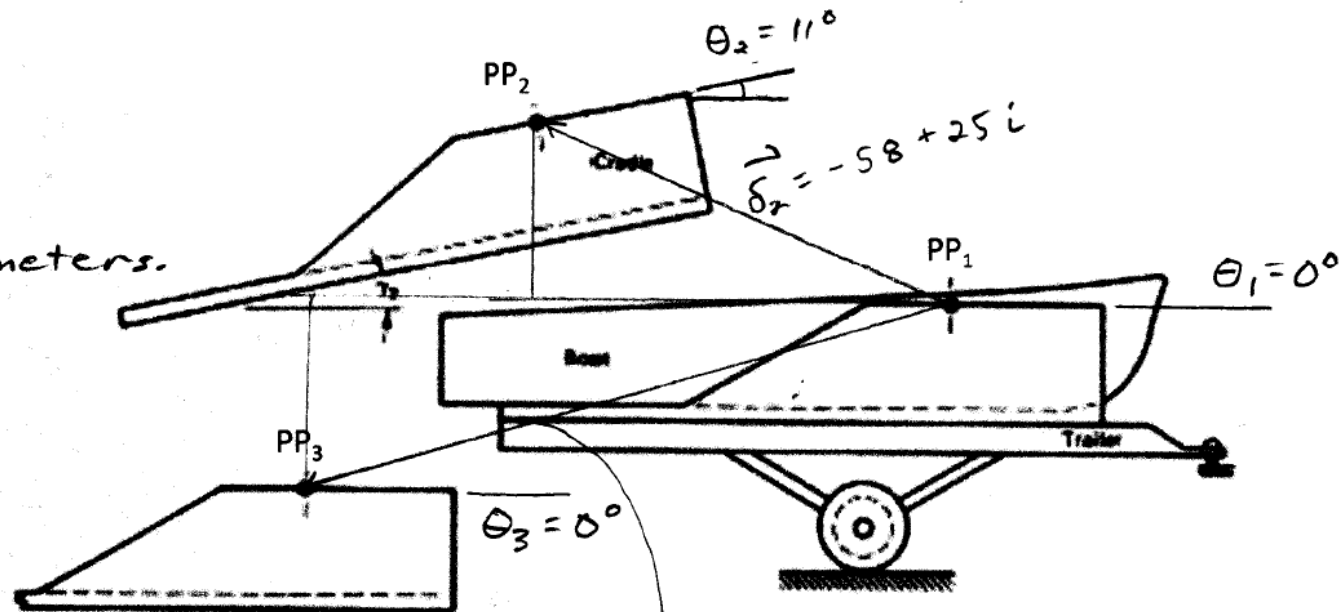
$$= 11^\circ - 0^\circ = 11^\circ$$

(+)

(+)

Imaginary
Real

All dimensions were
measured in millimeters.



$$\alpha_3 = \theta_3 - \theta_1$$

$$= 0^\circ - 0^\circ = 0^\circ$$

Example of a MATLAB program to solve for one dyad

% Dyad Synthesis -- Boat Example

% Assign a variable to convert degrees to radians

deg = pi / 180;

% define i

i = sqrt(-1);

% Free choices (change in orientation of links attached to ground)

beta2 = 50 * deg;

beta3 = 200 * deg;

% Knowns (change in orientation and position of coupler link)

alpha2 = 11 * deg;

alpha3 = 0 * deg;

delta2 = -58 + 25*i;

delta3 = -89 - 26*i;

% Calculations (using Cramer's rule to solve matrix equations)

A = [delta2, exp(i*alpha2)-1; delta3, exp(i*alpha3)-1];

B = [exp(i*beta2)-1, delta2; exp(i*beta3)-1, delta3];

C = [exp(i*beta2)-1, exp(i*alpha2)-1; exp(i*beta3)-1, exp(i*alpha3)-1];

% And the final answer is... (don't forget to repeat for the other dyad, by simply choosing different betas)

W = det(A) / det(C);

Z = det(B) / det(C);

Convert to radians
(MATLAB works in radians)



Randomly chosen values
of β_2 and β_3 (*Free choices*)



Analytical Method: Motion Generation with 3 Prescribed Positions

Problem Formulation:

$$\mathbf{W}(e^{i\beta_2} - 1) + \mathbf{Z}(e^{i\alpha_2} - 1) = \delta_2$$

$$\mathbf{W}(e^{i\beta_3} - 1) + \mathbf{Z}(e^{i\alpha_3} - 1) = \delta_3$$

Solution:
(using
Cramer's Rule)

$$\mathbf{W} = \frac{\begin{vmatrix} \delta_2 & e^{i\alpha_2} - 1 \\ \delta_3 & e^{i\alpha_3} - 1 \end{vmatrix}}{\begin{vmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \end{vmatrix}}$$

$$\mathbf{Z} = \frac{\begin{vmatrix} e^{i\beta_2} - 1 & \delta_2 \\ e^{i\beta_3} - 1 & \delta_3 \end{vmatrix}}{\begin{vmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \end{vmatrix}}$$

Results for Dyad A

(using the values of β shown below)

Chosen Values of Beta:

$$\text{beta2} = 50 * \text{deg}$$

$$\text{beta3} = 200 * \text{deg}$$

Matlab results -- W and Z vectors:

$$W = 46.7923 + 5.1534i$$

$$Z = -2.8087e+01 + 1.9838e+02i$$

Location of moving pivot (relative to PP_1):

$$A_1 = -Z = 28.1 - 198.4i$$

Location of ground pivot (relative to PP_1):

$$A_0 = A_1 - W = -18.7 - 203.6i$$

Draw these to see where they are.

Results for Dyad B

(using the values of β shown below)

Chosen Values of Beta:

$$\text{beta2} = 68 * \text{deg}$$

$$\text{beta3} = 114 * \text{deg}$$

Matlab results -- W and Z vectors:

$$W = 36.0577 + 41.8986i$$

$$Z = 90.5805 - 26.5302i$$

Location of moving pivot (relative to PP_1):

$$B_1 = -Z = -90.6 + 26.5i$$

Location of ground pivot (relative to PP_1):

$$B_0 = B_1 - W = -126.6 - 15.4i$$

Draw these to see where they are.

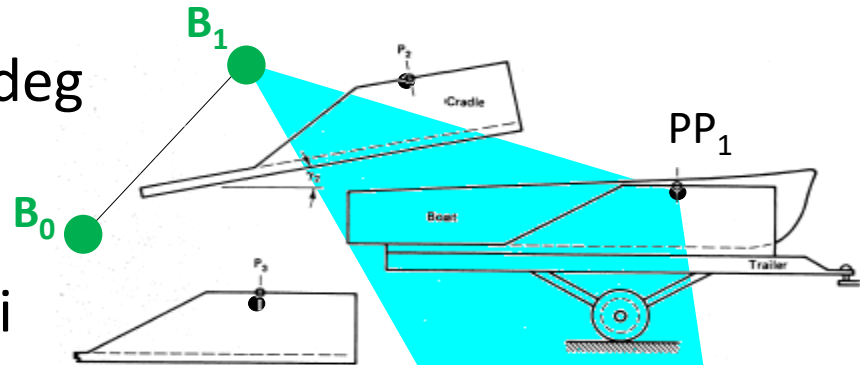
Solution using our chosen beta values

Using beta2=68deg, beta3=114deg

Dyad B:

$$B_1 = -Z = -90.6 + 26.5i$$

$$B_0 = B_1 - W = -126.6 - 15.4i$$



Notice how these are drawn using PP_1 as the origin.

Using beta2=50 deg, beta3=200deg

Dyad A:

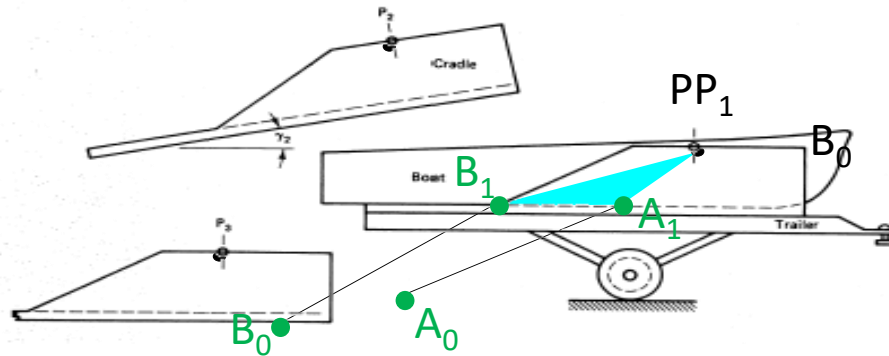
$$A_1 = -Z = 28.1 - 198.4i$$

$$A_0 = A_1 - W = -18.7 - 203.6i$$



Mechanism is too large. Choose different beta values.

Redone, iterating on beta values



Using $\beta_2=71.11\text{deg}$, $\beta_3=152\text{deg}$

Dyad B:

$$B_1 = -Z = -41.3 - 24.1i$$

$$B_0 = B_1 - W = -56.6 - 60.9i$$

Using $\beta_2=77.36\text{deg}$, $\beta_3=155.04\text{deg}$

Dyad A:

$$A_1 = -Z = -13.2 - 15.5i$$

$$A_0 = A_1 - W = -54.8 - 38.3i$$

Mechanism is much better